

IMAT MOCK EXAM 11: Last Rehearsal

60 questions · 100 minutes · scoring: +1.5 correct, −0.4 wrong, 0 blank · personal target: 42/90.0 · Mock exam · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Reading Skills, General Knowledge & Logical Reasoning (Q1-9)

1. "Take this medicine at evenly spaced times as prescribed and complete the full course, even if your symptoms improve, because stopping early can let the infection return. If you forget a dose, take it as soon as you remember, but skip the missed dose if your next dose is due soon; never take a double dose. Return any unused tablets to a pharmacy: they must not be kept to treat future illnesses or given to other people."

From the informational leaflet for an antibiotic

Which of the following statements is the correct interpretation of the instructions?

- A)** Stopping the course when symptoms improve is the recommended way to avoid side effects.
- B)** Leftover tablets must not be saved to treat a later illness.
- C)** Two tablets should be taken together whenever a dose is missed.
- D)** This antibiotic should always be taken with food.
- E)** Sharing the medicine with a family member who has similar symptoms is acceptable.

Answer: B

The leaflet states that unused tablets must be returned to a pharmacy and must not be kept to treat future illnesses or given to other people, so the statement about not saving leftover tablets is the correct interpretation. Taking two tablets after a missed dose contradicts the instruction never to take a double dose, and stopping when symptoms improve contradicts completing the full course. Nothing in the passage mentions taking the antibiotic with food, and it explicitly forbids giving the medicine to other people, family members included.

2. "In a large Spanish trial, volunteers at high cardiovascular risk were assigned by lottery to either a Mediterranean-style diet supplemented with extra-virgin olive oil and nuts, or to a control diet advised only to limit fat. Over about five years, the Mediterranean groups suffered roughly 30% fewer heart attacks, strokes and deaths from cardiovascular causes than the control group. The benefit appeared without meaningful weight loss and without a rise in calorie intake. The researchers note that the diet as a whole was tested, not any single ingredient, and that participants lived in a Mediterranean country where these foods are familiar and affordable."

From a newspaper report on nutrition research

Indicate which of the following statements is not correct:

- A) The benefit did not depend on the participants losing weight.
- B) Major cardiovascular events were about 30% lower in the Mediterranean groups.
- C) The researchers evaluated a whole dietary pattern rather than one ingredient.
- D) The trial demonstrated the protective effect of extra-virgin olive oil alone.
- E) Assignment to the Mediterranean or the control diet in the trial was random.

Answer: D

The report says the trial tested the Mediterranean diet as a whole, not any single ingredient, so claiming that it demonstrated a protective effect of extra-virgin olive oil alone is the statement that is not correct. All the other claims restate the passage: assignment was by lottery, so random, cardiovascular events fell by roughly 30%, the benefit appeared without meaningful weight loss, and a whole dietary pattern rather than one component was evaluated. Attributing the effect to a single ingredient goes beyond what a whole-diet trial can show.

3. "With the telescope I have detected four small stars that have escaped the notice of every astronomer before me. They attend a certain planet, arranged in a straight line with it, and their distances from the planet change: at times one hides behind the planet or passes in front of it, and they complete their circuits in unequal periods, the nearest the fastest."

Adapted from Galileo Galilei, Sidereus Nuncius (1610)

What did the author observe?

- A) Four moons orbiting a planet
- B) Spots on the surface of the Sun
- C) Comets passing close to the Sun
- D) Fixed stars that never change position
- E) The rings of Saturn

Answer: A

The four objects attend a planet, line up with it, hide behind it or pass in front of it, and complete circuits at unequal periods with the nearest fastest: bodies orbiting a planet. These are the four large moons of Jupiter, Galileo's discovery of 1610. Sunspots sit on the Sun rather than orbiting it, comets orbit the Sun, genuinely fixed stars show no such motion, and the rings of Saturn are not four separate bodies that repeatedly disappear behind a planet in a line with it.

4. "An antibiotic kills susceptible bacteria but spares any cells that already carry resistance genes, for example a gene encoding an enzyme that destroys the drug. The surviving cells multiply, and resistance genes can also pass between bacteria on small circles of DNA. In environments where antibiotics are used often, such as hospitals, resistant strains therefore make up a growing share of the bacterial population over time."

From a microbiology textbook

Only one of the following statements is coherent with the excerpt:

- A) Antibiotics create resistance genes inside susceptible bacteria.
- B) Susceptible and resistant cells are killed by an antibiotic to the same degree.
- C) Bacteria deliberately mutate when an antibiotic is present so that they can survive it.
- D) Resistance genes can never move from one bacterium to another.
- E) Frequent use of antibiotics favours the spread of resistant strains.

Answer: E

Because the antibiotic kills susceptible cells and spares those already carrying resistance genes, frequent use, as in hospitals, selects for the resistant strains, which then multiply and share genes on small DNA circles; so the statement that frequent antibiotic use favours the spread of resistant strains is coherent with the excerpt. Selection needs no purpose, so bacteria do not deliberately mutate to survive, and the drug does not create resistance genes, it only favours cells that already have them. The excerpt says the genes do pass between bacteria, and susceptible cells are by definition killed more than resistant ones.

5. Consider this statement: All squares are rectangles.

Which of the following statements is deducible from the given statement?

- A) All rectangles are squares.
- B) Some rectangles are not squares.
- C) If a figure is not a rectangle, then it is not a square.
- D) Every four-sided figure is a rectangle.
- E) Squares and rectangles share no geometric properties.

Answer: C

All squares are rectangles places squares inside the class of rectangles, so the only deducible statement is the contrapositive: a figure that is not a rectangle cannot be a square. Claiming that all rectangles are squares is the invalid converse, reversing an inclusion that need not hold both ways. Although some rectangles really are not squares in ordinary geometry, that fact does not follow from the given statement alone, which is compatible with every rectangle being a square, so it is the tempting but undeduced choice.

6. From the given statements:

If it rains this afternoon, the tennis match will be moved indoors.

If the match is moved indoors, it will start at 2 pm.

The match did not start at 2 pm.

Which conclusion can be deduced?

- A)** The match started late.
- B)** It did not rain this afternoon.
- C)** The match was cancelled.
- D)** The match was moved indoors but started at a different time.
- E)** It rained this afternoon.

Answer: B

Step 1: from the pair, if the match is moved indoors it starts at 2 pm, and the match did not start at 2 pm, modus tollens gives that the match was not moved indoors. Step 2: from the pair, if it rains the match is moved indoors, and it was not moved indoors, modus tollens again gives that it did not rain this afternoon. The premises say nothing about lateness, cancellation, or an indoor start at a different time, so those conclusions add unsupported information.

7. Amara, Bruno and Chiara live on three different floors of a small building, one person per floor. Bruno lives on the third floor, Amara lives directly above Bruno, and Chiara lives directly above Amara. On which floor does Chiara live?

- A)** On the sixth floor.
- B)** On the fourth floor.
- C)** On the third floor.
- D)** On the second floor.
- E)** On the fifth floor.

Answer: E

Bruno is fixed on the third floor. A person living directly above Bruno is on the fourth floor, so Amara is on the fourth; Chiara lives directly above Amara, so Chiara is on the fifth. The constraints determine all three floors uniquely, and any floor at or below the fourth would put Chiara at or under the position her condition forces her to be above.

8. Three students, Sara, Tom and Leo, each study exactly one foreign language among French, German and Spanish. Sara studies neither French nor German, and Leo studies German. Which of the following is correct?

- A)** Tom studies French.
- B)** Tom studies German.
- C)** Tom studies Spanish.
- D)** Sara studies French.
- E)** Leo studies Spanish.

Answer: A

Each student takes exactly one language. Sara rejects both French and German, so she must take Spanish; Leo already takes German; that leaves only French for Tom. Every other claim assigns Tom or Sara a language already taken or explicitly excluded, so the only correct statement is that Tom studies French.

9. What is the angle between the hour hand and the minute hand of an analogue clock at exactly 9:30?

- A) 75° B) 90° C) 105° D) 120° E) 180°

Answer: C

Step 1: at 30 minutes the minute hand points at 6, that is 180 degrees clockwise from 12. Step 2: the hour hand moves 30 degrees per hour, so at 9:30 it sits at $9.5 \times 30 = 285$ degrees, halfway between the 9 and the 10. Step 3: the angle between the hands is $285 - 180 = 105$ degrees. The value 90 degrees would hold only if the hour hand were still exactly on the 9; after half an hour it has drifted 15 degrees past it.

Section B - Biology (Q10-32)

10. The small amount of ATP that glycolysis itself produces is formed by:

- A) oxidation of pyruvate to acetyl-CoA.
B) hydrolysis of the NADH produced in the pathway.
C) chemiosmosis at the inner mitochondrial membrane.
D) direct transfer of a phosphate group from a phosphorylated intermediate to ADP.
E) the electron transport chain.

Answer: D

Glycolysis makes its small ATP yield by substrate-level phosphorylation: phosphorylated intermediates such as 1,3-bisphosphoglycerate and phosphoenolpyruvate transfer a phosphate group straight to ADP. Chemiosmosis at the inner mitochondrial membrane and the electron transport chain belong to the later stage of oxidative phosphorylation, oxidising pyruvate to acetyl-CoA yields no ATP, and the NADH made in glycolysis is cashed in at the chain, not by hydrolysis.

11. The reaction that converts pyruvate into acetyl-CoA before entry into the Krebs cycle takes place in:

- A) the cytosol.
B) the intermembrane space.
C) the Golgi apparatus.
D) the inner mitochondrial membrane.
E) the mitochondrial matrix.

Answer: E

Pyruvate made by cytosolic glycolysis is transported across the inner mitochondrial membrane into the matrix, where the pyruvate dehydrogenase complex oxidatively decarboxylates it to acetyl-CoA. The link reaction therefore occurs in the mitochondrial matrix, not in the cytosol where glycolysis runs, not in the intermembrane space where protons accumulate, and not at the inner membrane where the electron transport chain sits. The Golgi apparatus plays no part in respiration.

12. During the complete oxidation of one glucose molecule to CO₂ and water, carbon dioxide is released during:

- A) glycolysis and oxidative phosphorylation.
B) glycolysis and the Krebs cycle.
C) the link reaction and the Krebs cycle.
D) the Krebs cycle and oxidative phosphorylation.
E) glycolysis and the link reaction.

Answer: C

Step 1: glycolysis splits glucose into two pyruvates and releases no carbon dioxide. Step 2: each pyruvate loses one carbon as CO₂ on conversion to acetyl-CoA in the link reaction, 2 CO₂ in total. Step 3: two turns of the Krebs cycle, one per acetyl group, release the remaining 4 CO₂. Oxidative phosphorylation merely passes electrons to oxygen and forms water, so carbon dioxide appears only in the link reaction and the Krebs cycle.

13. A substance makes the inner mitochondrial membrane leaky to protons while leaving the electron transport complexes untouched. The most direct consequences are that:

- A)** electron transport stops while ATP synthesis continues.
- B)** electron transport accelerates while ATP synthesis falls.
- C)** ATP synthesis increases while electron transport slows.
- D)** the proton gradient grows and ATP synthesis accelerates.
- E)** electron transport and ATP synthesis both stop.

Answer: B

A leak that lets protons cross the inner membrane without passing through ATP synthase discharges the proton gradient, so the driving force for ATP synthesis collapses and ATP output falls. With the gradient no longer holding the electron chain back, electron transport speeds up rather than stopping. Both processes stopping would require poisoning the complexes themselves, which the stem explicitly rules out; this is precisely how uncouplers behave.

14. In the mitochondrial respiratory chain, electrons are carried from complex III to complex IV by:

- A)** cytochrome c. **B)** ubiquinone. **C)** oxygen. **D)** NAD^+ . **E)** ATP synthase.

Answer: A

In the chain, complexes I and II pass electrons to ubiquinone, which delivers them to complex III; from complex III the electrons travel to complex IV on cytochrome c, a small mobile protein on the outer face of the inner membrane, and oxygen finally accepts them at complex IV. Ubiquinone therefore works before complex III, oxygen is the terminal acceptor rather than a shuttle, NAD^+ is not a respiratory-chain carrier, and ATP synthase is powered by protons rather than carrying electrons.

15. In haemoglobin, each oxygen molecule binds to:

- A)** a molecule of bicarbonate inside the cell.
- B)** the membrane of the red blood cell.
- C)** a terminal amino group of a globin chain.
- D)** the iron ion at the centre of a heme group.
- E)** a hydrophobic pocket between two subunits.

Answer: D

Each of the four globin subunits of haemoglobin carries a heme group, and the iron ion at its centre binds one oxygen molecule reversibly, giving four O_2 per haemoglobin molecule. Carbon dioxide instead travels mainly as bicarbonate and partly on the terminal amino groups of the globin chains, and neither the red cell membrane, nor a hydrophobic pocket between subunits, nor a bicarbonate ion is the oxygen-binding site.

16. The largest fraction of the carbon dioxide carried in venous blood is transported:

- A)** as bicarbonate ions in the plasma.
- B)** inside the organelles of red blood cells.
- C)** as carbamino compounds on plasma proteins.
- D)** dissolved directly in the plasma.
- E)** bound to the iron atoms of haemoglobin.

Answer: A

Carbonic anhydrase inside red blood cells converts most incoming CO_2 to carbonic acid, which dissociates to bicarbonate; the bicarbonate then leaves the cell for the plasma in exchange for chloride, and about 70% of the CO_2 travels this way, by far the largest share. Only around a fifth binds to haemoglobin as carbamino compounds and a few percent dissolves directly. The heme iron binds oxygen, not carbon dioxide, and mature red blood cells contain no organelles at all.

17. Compared with fast-twitch fibres, slow-twitch (type I) muscle fibres:

- A)** contain fewer mitochondria and fatigue rapidly.
- B)** rely mainly on anaerobic glycolysis for their ATP.
- C)** contain more mitochondria and are more resistant to fatigue.
- D)** contract more rapidly but for shorter periods.
- E)** store no myoglobin at all.

Answer: C

Slow-twitch type I fibres are endurance fibres packed with mitochondria and myoglobin and richly supplied with capillaries, so their oxidative metabolism lets them contract for long periods without fatiguing. Fast-twitch fibres are the ones with fewer mitochondria that rely mainly on anaerobic glycolysis and tire rapidly. Slow fibres are far from lacking myoglobin; their high myoglobin content is what makes them red.

18. The thin layer of cells lining the inner surface of the heart chambers and of all blood vessels is called the:

- A)** mesothelium.
- B)** endothelium.
- C)** periosteum.
- D)** collagen sheath.
- E)** perichondrium.

Answer: B

The simple squamous epithelium that lines the heart chambers and every blood vessel is called the endothelium. Mesothelium instead lines the closed body cavities such as the pleural and pericardial spaces, perichondrium wraps cartilage, periosteum wraps bone, and a collagen sheath is extracellular connective tissue rather than a cellular lining.

19. The activity of an enzyme rises with temperature up to an optimum and then falls steeply at higher temperatures because the enzyme:

- A)** switches to catalysing a different reaction.
- B)** has run out of substrate.
- C)** is broken down into its component fatty acids.
- D)** starts to synthesise an inhibitor of itself.
- E)** begins to denature, losing its functional three-dimensional shape.

Answer: E

Above the optimum temperature, extra heat breaks the weak hydrogen and ionic interactions that hold the enzyme's three-dimensional fold, so the protein denatures and the active site loses its shape, and activity falls steeply. Substrate supply does not change with temperature, enzymes are proteins that break down to amino acids rather than fatty acids, and no enzyme switches reaction or manufactures its own inhibitor on heating.

- 20.** Inhibition by a non-competitive inhibitor differs from competitive inhibition in that it:
- A)** affects only the apparent affinity of the enzyme for its substrate.
 - B)** is reversed completely by adding excess substrate.
 - C)** disappears when the reaction mixture is cooled.
 - D)** lowers the maximum rate of the reaction and is not overcome by excess substrate.
 - E)** involves the inhibitor binding at the active site.

Answer: D

A non-competitive inhibitor binds at an allosteric site away from the active site and inactivates those enzyme molecules completely, so the maximum reaction rate falls and no excess of substrate can restore it. Being fully reversed by excess substrate, and changing only the apparent affinity for substrate, are the defining behaviours of competitive inhibition, which occupies the active site itself. Temperature has nothing specific to do with the distinction.

- 21.** Between meals, bile is stored and concentrated in the:
- A)** liver.
 - B)** stomach.
 - C)** small intestine.
 - D)** pancreas.
 - E)** gall bladder.

Answer: E

The liver produces bile continuously, but between meals the bile is diverted into the gall bladder, which stores it and concentrates it by reabsorbing water until a fatty meal triggers its contraction into the duodenum. The pancreas produces digestive enzymes and bicarbonate but no bile, and the small intestine is where bile acts, not where it is stored between meals.

- 22.** Soon after a meal rich in carbohydrates, the liver responds to the prevailing hormonal signal by:
- A)** converting glucose into glycogen for storage.
 - B)** converting excess glucose into urea.
 - C)** breaking glycogen down into free glucose.
 - D)** increasing the loss of glucose in the bile.
 - E)** secreting glucagon into the blood.

Answer: A

A carbohydrate-rich meal raises blood glucose, which stimulates insulin secretion, and insulin makes the liver polymerise glucose into glycogen for storage, glycogenesis. Breaking glycogen down to free glucose is the opposite, the fasting response driven by glucagon, whose secretion falls at this time; glucagon is released by the pancreas, not the liver; and glucose contains no nitrogen, so it cannot be turned into urea.

23. The osmotic gradient of the renal medulla, which allows the production of concentrated urine, is created mainly by active salt transport in the:

- A)** proximal convoluted tubule.
- B)** distal convoluted tubule.
- C)** loop of Henle.
- D)** glomerulus.
- E)** renal artery.

Answer: C

The thick ascending limb of the loop of Henle actively pumps sodium and chloride into the medullary tissue while remaining impermeable to water, creating the osmotic gradient that the collecting duct, under ADH, later uses to reabsorb water and produce concentrated urine. The glomerulus only filters blood, the proximal tubule reabsorbs glucose and most solutes without building the gradient, the distal tubule fine-tunes ion balance, and a blood vessel cannot generate a tissue gradient.

24. The cells whose beating cilia sweep mucus and trapped dust upward out of the airways are:

- A)** alveolar macrophages.
- B)** ciliated epithelial cells.
- C)** chondrocytes.
- D)** type I alveolar cells.
- E)** goblet cells.

Answer: B

Ciliated epithelial cells line the conducting airways, and their coordinated ciliary beating transports the overlying mucus blanket with its trapped dust up to the pharynx, where it is swallowed, the mucociliary escalator. Goblet cells only secrete the mucus, alveolar macrophages patrol the alveoli deep in the lung, type I alveolar cells form the gas-exchange wall, and chondrocytes build the cartilage rings.

25. Iodine is required in the human diet mainly for the synthesis of:

- A)** cortisol.
- B)** adrenaline.
- C)** antidiuretic hormone.
- D)** thyroxine.
- E)** insulin.

Answer: D

The thyroid is the only tissue that concentrates dietary iodide, which it incorporates into thyroxine (T4) and triiodothyronine (T3); chronic iodine shortage therefore enlarges the gland into a goitre. Insulin, adrenaline, antidiuretic hormone and cortisol contain no iodine at all, so none of them can be the answer.

26. The α -helix and the β -pleated sheet are features of which level of protein structure?

- A) Primary structure
- B) Secondary structure
- C) The prosthetic group level
- D) Tertiary structure
- E) Quaternary structure

Answer: B

The alpha-helix and the beta-pleated sheet are regular local folds of the polypeptide backbone stabilised by hydrogen bonds between backbone groups, which is exactly the definition of secondary structure. Primary structure is the amino acid sequence, tertiary structure is the complete three-dimensional fold of one chain, quaternary structure is the assembly of several chains, and no prosthetic-group level exists in the standard scheme.

27. According to the fluid mosaic model of the plasma membrane, the proteins of the membrane:

- A) can drift laterally within the lipid bilayer.
- B) are composed of carbohydrate.
- C) form a solid continuous coat over the bilayer.
- D) occur only on the outer surface of the membrane.
- E) are each fixed rigidly in a single position.

Answer: A

In the fluid mosaic model the bilayer is a two-dimensional fluid in which both lipids and integral proteins can drift laterally, classically shown when fluorescently labelled proteins of a fused mouse and human cell intermixed across the hybrid membrane. The proteins are not rigidly fixed in place, they span the membrane or sit on both faces rather than only the outer surface, they are proteins rather than carbohydrate, and the membrane is a mosaic, not a solid continuous coat.

28. The digestive enzymes inside lysosomes have an optimum pH of about:

- A) 2. B) 5. C) 7.4. D) 9. E) 12.

Answer: B

Lysosomal hydrolases have their optimum near pH 5, maintained by proton pumps that acidify the organelle's interior. A pH near 2 is typical of the stomach lumen, 7.4 of blood, and 9 or 12 are far too alkaline. The mildly acidic optimum is also a safety feature: any hydrolase escaping into the cytosol at about pH 7.2 is largely inactivated.

29. Catalase, the enzyme that decomposes hydrogen peroxide into water and oxygen, is concentrated in which organelle?

- A) The Golgi apparatus
- B) The mitochondrion
- C) The nucleolus
- D) The lysosome
- E) The peroxisome

Answer: E

Catalase is the signature enzyme of the peroxisome, an organelle in which oxidase reactions produce hydrogen peroxide and catalase promptly decomposes it to water and oxygen. Lysosomes instead digest macromolecules at acidic pH, mitochondria make ATP, the Golgi apparatus modifies and packages proteins, and the nucleolus assembles ribosomal subunits.

30. For a cross between two heterozygotes, $Aa \times Aa$, the expected ratio of genotypes $AA : Aa : aa$ among the offspring is:

- A)** 1:1:1 **B)** 1:2:1 **C)** 2:1:1 **D)** 3:1 **E)** 9:3:3:1

Answer: B

A Punnett square for $Aa \times Aa$ gives the four equally likely combinations AA , Aa , Aa and aa , so the genotype ratio is 1:2:1. The familiar 3:1 ratio is phenotypic, arising because AA and Aa look identical when the dominant allele masks the recessive one; 1:1:1 comes from crossing a heterozygote with a homozygous recessive, and 9:3:3:1 belongs to a dihybrid cross.

31. Red-green colour blindness is caused by a recessive allele carried on the X chromosome. A man with colour blindness and a woman with normal vision whose father was colour blind have children. Which statement is correct?

- A)** All of the children will have normal vision.
B) All of the daughters will be colour blind.
C) On average, half of the sons and half of the daughters will be colour blind.
D) All of the sons will be colour blind.
E) The daughters cannot have received the allele from their father.

Answer: C

Let c be the recessive colour-blind allele on the X chromosome. The mother must be a carrier, because her colour-blind father passed her his only X chromosome, the one bearing c ; the father is colour blind, so his X carries c and his Y goes to sons. Step 1: every daughter receives the father's c -bearing X, and from her carrier mother receives a normal X or a c -bearing X with equal chance, so half the daughters are unaffected carriers and half are colour blind. Step 2: every son receives the father's Y and one maternal X, so half the sons are colour blind and half have normal vision. Hence on average half of each sex is colour blind; the claim that daughters cannot have received the allele from their father is exactly backwards, since all of them did, and the claim that all daughters are colour blind ignores the normal allele half of them get from their mother.

32. During elongation of translation, the new peptide bond forms between the growing polypeptide chain and:

- A)** the anticodon of the following tRNA.
B) the ribosomal RNA of the small subunit.
C) the tRNA about to leave through the E site.
D) the amino acid carried by the tRNA that has just entered the A site.
E) the codon currently being read in the mRNA.

Answer: D

During elongation the peptidyl transferase activity of the large ribosomal subunit transfers the growing polypeptide from the tRNA in the P site onto the amino acid held by the tRNA that has just entered the A site, so each new peptide bond forms at that newly arrived aminoacyl-tRNA. Codons and anticodons carry information but no amino acid, the E-site tRNA leaves only after it has already surrendered its amino acid, and ribosomal RNA is the catalytic scaffold rather than a bond partner.

Section C - Chemistry (Q33-47)

33. What is the pH of a 0.001 mol/dm^3 aqueous solution of sodium hydroxide, a strong base? ($K_w = 1.0 \times 10^{-14}$ at 25°C)

- A) 3.0 B) 7.0 C) 10.0 D) 11.0 E) 13.0

Answer: D

Sodium hydroxide is a strong base and dissociates completely, so the hydroxide concentration equals $1.0 \times 10^{-3} \text{ mol/dm}^3$ and $\text{pOH} = -\log(10^{-3}) = 3$.

Using $\text{pH} + \text{pOH} = 14$ at 25°C gives $\text{pH} = 14 - 3 = 11.0$

The value 3.0 is the pOH misread as pH, and 13.0 would require a ten times more concentrated, 0.1 mol/dm^3 , solution.

34. At 25°C , pure water has $K_w = 1.0 \times 10^{-14}$ and pH 7. At 80°C , K_w rises to 4.0×10^{-13} . In pure water at 80°C , the concentration of H^+ and the nature of the water are:

- A) $[\text{H}^+] = 1.0 \times 10^{-7} \text{ mol/dm}^3$ and the water is neutral.
B) $[\text{H}^+] = 6.3 \times 10^{-7} \text{ mol/dm}^3$ and the water is acidic.
C) $[\text{H}^+] = 6.3 \times 10^{-7} \text{ mol/dm}^3$ and the water is neutral.
D) $[\text{H}^+] = 2.0 \times 10^{-13} \text{ mol/dm}^3$ and the water is basic.
E) $[\text{H}^+] = 4.0 \times 10^{-13} \text{ mol/dm}^3$ and the water is acidic.

Answer: C

In pure water every ionisation event produces one H^+ and one OH^- , so the two concentrations stay equal and each equals the square root of K_w : $\sqrt{(4.0 \times 10^{-13})} = \sqrt{(40 \times 10^{-14})} \approx 6.3 \times 10^{-7} \text{ mol/dm}^3$. Equal concentrations of H^+ and OH^- is precisely what neutral means, even though the pH is now about 6.2 rather than 7, because neutrality is defined by the balance of the two ions and not by the number 7. Setting the H^+ concentration equal to K_w itself confuses the ion product with an ion concentration, and keeping 10^{-7} ignores the new temperature.

35. Which of the following substances has a giant covalent (network covalent) structure?

- A) Diamond
B) Sodium chloride
C) Methane
D) Ice
E) Copper

Answer: A

Diamond is one continuous lattice in which every carbon atom is covalently bonded to four neighbours throughout the crystal, the defining giant covalent network. Sodium chloride is a giant ionic lattice, copper is a giant metallic lattice, and ice and methane consist of discrete molecules: ice molecules cling to each other only by hydrogen bonds and methane molecules only by weak dispersion forces.

36. Water ($M = 18 \text{ g/mol}$) boils at 100°C , while methane ($M = 16 \text{ g/mol}$) boils at -162°C . The main reason for the much higher boiling point of water is:

- A)** ionic bonds between water molecules.
- B)** the larger number of covalent bonds inside each water molecule.
- C)** the slightly greater molecular mass of water.
- D)** covalent bonds linking the atoms of different molecules.
- E)** hydrogen bonds between water molecules.

Answer: E

Each water molecule can both donate and accept hydrogen bonds through its O-H groups and oxygen lone pairs, and this strong intermolecular attraction must be overcome to vaporise the liquid, hence the boiling point of 100°C . There are no ionic bonds between water molecules, and the covalent bonds inside each molecule are not broken on boiling. A mass difference of just 2 g/mol cannot explain a gap of about 260 degrees, since methane's mass gives it only weak dispersion forces.

37. What is the oxidation number of phosphorus in H_3PO_4 ?

- A)** +1 **B)** +3 **C)** +5 **D)** +6 **E)** +7

Answer: C

Assign the usual oxidation numbers, hydrogen +1 and oxygen -2.

For the neutral molecule: $3 \times (+1) + x + 4 \times (-2) = 0$, so $x - 5 = 0$ and $x = +5$

The value +3 belongs to phosphorous acid, H_3PO_3 , which has one oxygen fewer, and +6 or +7 cannot be balanced by this formula at all.

38. In the reaction $2\text{FeCl}_2 + \text{Cl}_2 \rightarrow 2\text{FeCl}_3$, which species is the oxidising agent?

- A)** Fe^{2+} , because it combines with chlorine.
- B)** Fe^{3+} , because it is the product of the reaction.
- C)** Cl^- , because it carries a negative charge.
- D)** Neither species, because no electrons are transferred.
- E)** Cl_2 , because it accepts electrons from the iron and is itself reduced.

Answer: E

Iron rises from +2 in FeCl_2 to +3 in FeCl_3 , so the iron loses electrons and is oxidised; chlorine falls from 0 in Cl_2 to -1 in the product, so chlorine gains electrons and is reduced. The species that accepts electrons is by definition the oxidising agent, so Cl_2 plays that role and the iron(II) is the reducing agent. Carrying a negative charge, or merely being the product, does not determine the role.

39. How many open-chain alcohols have the molecular formula $\text{C}_4\text{H}_{10}\text{O}$?

- A)** 2 **B)** 3 **C)** 4 **D)** 5 **E)** 6

Answer: C

Step 1: C_4H_{10} has only two carbon skeletons, straight-chain butane and branched 2-methylpropane.
Step 2: on butane the hydroxyl can sit on carbon 1 or carbon 2, since positions 3 and 4 mirror 1 and 2, giving butan-1-ol and butan-2-ol. Step 3: on 2-methylpropane the hydroxyl on an outer carbon gives 2-methylpropan-1-ol and on the central carbon gives 2-methylpropan-2-ol. That totals 4 alcohols; the open-chain ethers with this same formula are correctly excluded because the question asks only for alcohols.

40. The bond formed between the amino group of one amino acid and the carboxyl group of another is called:

- A)** a phosphodiester bond.
- B)** an ester bond.
- C)** a glycosidic bond.
- D)** a peptide bond.
- E)** a hydrogen bond.

Answer: D

The condensation reaction between the carboxyl group of one amino acid and the amino group of another releases water and forms the -CO-NH- linkage called a peptide bond. Glycosidic bonds join monosaccharides, ester bonds join glycerol to fatty acids, phosphodiester bonds join nucleotides in nucleic acids, and a hydrogen bond is a weak non-covalent attraction rather than a true covalent linkage.

41. What amount, in moles, is present in 8.8 g of carbon dioxide? ($M(\text{CO}_2) = 44 \text{ g/mol}$)

- A)** 0.1 mol **B)** 0.2 mol **C)** 0.44 mol **D)** 2.0 mol **E)** 4.4 mol

Answer: B

Amount of substance = mass $\div$ molar mass = $8.8 \text{ g} \div 44 \text{ g/mol} = 0.2 \text{ mol}$, since $44 \times 0.2 = 8.8$. The value 0.44 comes from dividing by 20 instead of 44, and values above 1 mol come from inverting the division and multiplying mass by molar mass.

42. A sample of gas occupies 4.0 dm^3 at 300 K and 100 kPa. If the sample is heated to 600 K at constant pressure, its volume becomes:

- A)** 2.0 dm^3 **B)** 4.0 dm^3 **C)** 6.0 dm^3 **D)** 8.0 dm^3 **E)** 12.0 dm^3

Answer: D

Charles's law at constant pressure: $V_1/T_1 = V_2/T_2$

Substituting, $V_2 = 4.0 \text{ dm}^3 \times (600 \text{ K} \div 300 \text{ K}) = 4.0 \times 2 = 8.0 \text{ dm}^3$, because doubling the absolute temperature doubles the volume. Halving the volume to 2.0 dm^3 reverses the law, as if heating shrank the gas.

43. Going down group 1 of the periodic table, from lithium towards francium, the reactivity of the metals with water:

- A)** increases, because the outer electron is further from the nucleus and is lost more easily.
- B)** remains constant, because every group 1 metal has one outer electron.
- C)** decreases, because the outer electron is held more tightly.
- D)** decreases, because the atoms become smaller.
- E)** first increases and then decreases.

Answer: A

Going down group 1 each element adds a shell, so the atoms grow larger and the single outer electron sits further from the nucleus with more shielding between them; the attraction holding it weakens, it is lost more easily, and reactivity with water increases from lithium towards caesium. The atoms do not shrink down the group, and having one outer electron in common does not make reactivity constant, because what matters is how easily that electron escapes.

44. The electron configuration of a neutral atom of sulfur ($Z = 16$) is:

- A) $1s^2 2s^2 2p^6 3s^2 3p^4$
- B) $1s^2 2s^2 2p^6 3s^2 3p^6$
- C) $1s^2 2s^2 2p^6 3s^1 3p^5$
- D) $1s^2 2s^2 2p^5 3s^2 3p^5$
- E) $1s^2 2s^2 2p^4 3s^2 3p^6$

Answer: A

Fill the sublevels in order for 16 electrons: $1s^2$ (2) + $2s^2$ (4) + $2p^6$ (10) + $3s^2$ (12), leaving $16 - 12 = 4$ electrons for 3p, so $1s^2 2s^2 2p^6 3s^2 3p^4$. The configuration ending $3p^6$ totals 18 electrons and belongs to argon, not sulfur; moving an electron from 3s to 3p, or draining 2p to top up 3p, keeps the same total but violates the sublevel filling order.

45. For the equilibrium $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$, the equilibrium constant K is given by:

- A) $K = 2[\text{SO}_3] / (2[\text{SO}_2] + [\text{O}_2])$
- B) $K = ([\text{SO}_2]^2[\text{O}_2]) / [\text{SO}_3]^2$
- C) $K = [\text{SO}_3] / ([\text{SO}_2]^2[\text{O}_2])$
- D) $K = [\text{SO}_3]^2 / ([\text{SO}_2][\text{O}_2])$
- E) $K = [\text{SO}_3]^2 / ([\text{SO}_2]^2[\text{O}_2])$

Answer: E

For an equilibrium $a\text{A} + b\text{B} \rightleftharpoons c\text{C}$, the constant is the product concentrations divided by the reactant concentrations, each raised to its coefficient in the balanced equation.

Here the coefficients are 2, 1 and 2, so $K = [\text{SO}_3]^2 \div ([\text{SO}_2]^2 \times [\text{O}_2])$

Dropping the square on the SO_2 concentration, inverting the whole fraction, or using the coefficients as multiplying factors instead of exponents all give invalid expressions.

46. When the temperature of a reaction mixture is raised, the rate of the reaction increases. Which property of the reaction remains unchanged?

- A) The rate at which products form.
- B) The number of reactant molecules able to react.
- C) The activation energy of the reaction.
- D) The frequency of collisions between reactant molecules.
- E) The fraction of molecular collisions energetic enough to react.

Answer: C

Heating reshapes the Maxwell-Boltzmann energy distribution so that a larger fraction of collisions exceeds the activation energy, and collisions also occur somewhat more frequently; both effects raise the rate. The activation energy itself is fixed by the reaction pathway in the energy profile and does not depend on temperature, so it is the property that stays unchanged unless a catalyst alters the mechanism. The rate is by definition what increases, and the fraction of sufficiently energetic collisions rises, so neither can be the unchanged quantity.

47. The compound formed between the aluminium ion, Al^{3+} , and the sulfate ion, SO_4^{2-} , has the formula:

- A) Al_2SO_4
- B) $\text{Al}_2(\text{SO}_4)_3$
- C) $\text{Al}(\text{SO}_4)_3$
- D) $\text{Al}_3(\text{SO}_4)_2$
- E) AlSO_4

Answer: B

The total positive charge must cancel the total negative charge: the least common multiple of 3 and 2 is 6, so two Al^{3+} ions give +6 and three SO_4^{2-} ions give -6, requiring $\text{Al}_2(\text{SO}_4)_3$, with parentheses because the sulfate ion appears three times. The alternative formulas leave a net charge, for example +1 for AlSO_4 and -3 for $\text{Al}(\text{SO}_4)_3$, so none of them can describe a neutral compound.

Section D - Mathematics (Q48-54)

48. Solve the inequality $1 - 2x \geq 5$.

- A) $x \leq -2$ B) $x \geq 2$ C) $x \geq -2$ D) $x \leq -1$ E) $x \leq 2$

Answer: A

Step 1: subtract 1 from both sides, giving $-2x \geq 4$. Step 2: divide both sides by -2 ; dividing by a negative number reverses the inequality sign, so $x \leq -2$. Forgetting the reversal gives $x \geq -2$, and mishandling the sign of the constant produces the versions ending at $+2$, so the sign flip is the whole point of the question.

49. The set of real solutions of the inequality $(x - 2)^2 \geq 0$ is:

- A) $x \geq 2$
B) only $x = 2$
C) the empty set
D) the set of all real numbers
E) $x \leq 2$

Answer: D

The square of any real number is never negative, and $(x - 2)^2$ equals zero only at $x = 2$ while being strictly positive everywhere else. The condition $(x - 2)^2 \geq 0$ is therefore satisfied by every real number, so the solution set is all real numbers. Restricting the answer to $x = 2$ solves the stricter equation $(x - 2)^2 = 0$, not the inequality.

50. In a class of 40 students, 25 study French, 18 study German and 9 study both languages. How many students study neither language?

- A) 6 B) 9 C) 15 D) 16 E) 34

Answer: A

By inclusion-exclusion, the number studying at least one language is $25 + 18 - 9 = 34$, because the 9 taking both were counted twice. The class has 40 students, so $40 - 34 = 6$ study neither language. The value 34 reports the union instead of the complement, and the value 15 merely counts students not taking French while forgetting the German-only students.

51. If $3x + 2y = 12$ and $y = 3$, then x equals:

- A) 1 B) 2 C) 3 D) 4 E) 6

Answer: B

Substitute $y = 3$ into $3x + 2y = 12$: $3x + 6 = 12$, so $3x = 6$ and $x = 2$

The answer 3 simply copies the given y , and dividing 12 by 3 before subtracting the 6 gives the incorrect 4.

52. The expression $x^2 - 16$ can be factorised as:

- A) $(x - 4)^2$ B) $(x + 4)^2$ C) $x(x - 16)$ D) $(x - 2)(x + 8)$ E) $(x - 4)(x + 4)$

Answer: E

Recognise the difference of squares: $x^2 - 16 = x^2 - 4^2 = (x - 4)(x + 4)$

Expanding back, $x^2 + 4x - 4x - 16 = x^2 - 16$, confirming the middle terms cancel. The perfect squares $(x - 4)^2$ and $(x + 4)^2$ each add an extra $8x$ term, $x(x - 16)$ expands to $x^2 - 16x$, and $(x - 2)(x + 8)$ expands to $x^2 + 6x - 16$.

53. A right-angled triangle has legs of 6 cm and 8 cm. Its perimeter is:

- A) 14 cm B) 20 cm C) 24 cm D) 26 cm E) 30 cm

Answer: C

Pythagoras' theorem gives the hypotenuse: $\sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$ cm

The perimeter is $6 + 8 + 10 = 24$ cm

Adding only the two legs gives 14 cm, which omits the longest side entirely.

54. In a right-angled triangle, one angle is 30° and the hypotenuse has length 10 cm. The side opposite the 30° angle has length:

- A) 2.5 cm B) 5 cm C) $5\sqrt{3}$ cm D) 10 cm E) 12 cm

Answer: B

The side opposite an angle relates to the hypotenuse by the sine: $\text{opposite} = \text{hypotenuse} \times \sin(30^\circ) = 10 \times 1/2 = 5$ cm. The value $5\sqrt{3}$ cm would be the side adjacent to the 30° angle, since $\cos(30^\circ) = \sqrt{3}/2$, and a leg of 10 cm is impossible because it would equal the hypotenuse of the same right triangle.

Section E - Physics (Q55-60)

55. The velocity-time graph of a cyclist is a horizontal line at 6 m/s for the whole interval from $t = 0$ to $t = 10$ s. The distance covered in this interval is:

- A) 0.6 m B) 6 m C) 30 m D) 60 m E) 600 m

Answer: D

Distance is the area under the velocity-time graph, here a rectangle of height 6 m/s and width 10 s: $6 \times 10 = 60$ m. The value 30 m would follow from using an average speed of 3 m/s, appropriate for uniform acceleration from rest but not for a constant velocity, and 600 m multiplies by 100 instead of 10.

56. When a 12 V battery is connected across a lamp, a current of 0.5 A flows. The resistance of the lamp is:

- A) 6 Ω B) 12 Ω C) 24 Ω D) 48 Ω E) 60 Ω

Answer: C

Ohm's law: $R = V \div I = 12 \text{ V} \div 0.5 \text{ A} = 24 \Omega$

Multiplying instead of dividing gives 6, which is actually the power in watts ($V \times I$), and doubling the voltage again yields the spurious 48 Ω .

57. A 60 W lamp is left switched on for 2 hours. The electrical energy it consumes is:

- A) 0.012 kWh B) 0.12 kWh C) 1.2 kWh D) 12 kWh E) 120 kWh

Answer: B

Energy = power $\times$ time.

Convert 60 W to 0.06 kW, then $0.06 \text{ kW} \times 2 \text{ h} = 0.12 \text{ kWh}$

In joules the same result is $60 \text{ W} \times 7200 \text{ s} = 432000 \text{ J}$, and $432000 \div 3600000 = 0.12 \text{ kWh}$; each neighbouring option merely shifts the decimal point.

58. A box slides across a level floor at constant velocity. The resultant (net) force acting on the box is:
- A) directed forwards and larger than the friction force.
 - B) equal to the friction force and directed forwards.
 - C) equal to the weight of the box.
 - D) directed backwards.
 - E) zero.

Answer: E

Constant velocity means zero acceleration, so Newton's second law with $a = 0$ forces the resultant force to be zero: the forward push exactly balances friction, and the weight is separately balanced by the normal reaction. Any non-zero resultant, whether forwards or backwards, would accelerate the box and change its velocity.

59. An astronaut weighs about one sixth as much standing on the Moon as standing on Earth because:
- A) the Moon's smaller gravitational acceleration acts on the astronaut's unchanged mass.
 - B) the astronaut's body contains less matter on the Moon.
 - C) the Moon has no atmosphere to push up on the astronaut.
 - D) weight cannot be measured where there is no water or air.
 - E) part of the astronaut's weight is still carried by the distant Earth.

Answer: A

Weight is mass times local gravitational acceleration, $W = mg$. The astronaut's mass is the same everywhere, but the Moon's smaller mass and radius give a surface gravity of about 1.6 m/s^2 , roughly one sixth of Earth's 9.8 m/s^2 , so the same body weighs one sixth as much. The amount of matter in the astronaut does not change with location, and the presence or absence of an atmosphere has no bearing on weight.

60. An object weighs 50 N in air. When it is fully submerged in water, its apparent weight is 40 N. The buoyant force exerted by the water on the object is:
- A) 5 N B) 10 N C) 40 N D) 50 N E) 90 N

Answer: B

The apparent weight equals the true weight minus the upward buoyant force, so $B = 50 - 40 = 10 \text{ N}$. By Archimedes' principle this equals the weight of the water displaced by the object. The values 40 N and 50 N are the apparent and true weights themselves, not the buoyant force, and 90 N adds the two weights where the physics subtracts them.